

Kochen–Specker uncolorability in root-of-unity coordinate alphabets requires exactly $6 \mid n$

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(Dated: March 2026)

We prove that the ray set S_n generated by the n th roots of unity in $\mathbb{C}^3$ is Kochen–Specker uncolorable if and only if $6 \mid n$. Sufficiency follows from the embedding of Cabello’s Eisenstein 33-vector KS set. Necessity is proved by algebraic analysis of vanishing sums among roots of unity: when n is odd and $3 \mid n$, a projective collapse argument shows every triad is pairwise disjoint (no two share a ray) and the KS coloring constraints decompose into independent subproblems; when $3 \nmid n$ and $2 \mid n$, the orthogonality graph on each coordinate plane is a perfect matching, admitting an explicit coloring. The factorization $6 = 2 \times 3$ mirrors the ring-theoretic result of Cortez, Morales, and Reyes that $\mathbb{Z}[1/6]$ is the minimal ring supporting KS contextuality in dimension 3, though the two results are logically independent: ours concerns colorability of a coordinate-alphabet family, theirs concerns realizability of projectors over subrings of $\mathbb{Q}$.

INTRODUCTION

The Kochen–Specker (KS) theorem [1] establishes that quantum measurements in dimension $d \geq 3$ cannot be explained by noncontextual hidden-variable theories, by exhibiting a finite set of rays admitting no consistent $\{0, 1\}$ -valued coloring. A natural source of KS constructions is the cyclotomic family: for each positive integer n , the alphabet $\{0\} \cup \{\zeta^k : k = 0, \dots, n-1\}$ with $\zeta = e^{2\pi i/n}$ generates a ray set S_n in $\mathbb{C}^3$ via all projectively distinct nonzero vectors with coordinates drawn from this alphabet.

Cabello [2] showed that S_6 contains a 33-vector KS set built from the Eisenstein integers $\mathbb{Z}[\omega]$ ($\omega = e^{2\pi i/3}$), which he identified as the “simplest” KS set by the criterion of fewest bases. Cortez, Morales, and Reyes [3] proved that $\mathbb{Z}[1/6]$ is the minimal ring extension of $\mathbb{Z}$ exhibiting KS contextuality in dimension 3—that is, the primes 2 and 3 are both necessary.

These results suggest a sharp characterization: KS-uncolorability in the root-of-unity coordinate family should require exactly $6 \mid n$. We prove this. (We emphasize that the relevant structure is the available root subgroup μ_n , not the ambient field: $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\zeta_6)$, yet S_3 is colorable while S_6 is not.)

Theorem 1. *The ray set S_n generated by the n th roots of unity in $\mathbb{C}^3$ is KS-uncolorable if and only if $6 \mid n$.*

PRELIMINARIES

A *ray* in $\mathbb{C}^3$ is a one-dimensional subspace, represented by a nonzero vector up to scalar multiples. Two rays v, w are *orthogonal* if $\langle v|w \rangle = \sum_k \bar{v}_k w_k = 0$. An orthogonal *triad* is a set of three mutually orthogonal rays (an orthonormal basis up to normalization). A *KS coloring* assigns each ray a value in $\{0, 1\}$ such that (I) orthogonal rays cannot both receive 1, and (II) every triad contains

exactly one 1. A ray set is *KS-uncolorable* if no such coloring exists.

For $\zeta = e^{2\pi i/n}$, the ray set S_n consists of all projectively distinct (modulo $\mathbb{C}^\times$) nonzero vectors with coordinates in $\{0\} \cup \{\zeta^j : 0 \leq j < n\}$. We classify rays by *zero pattern*: *axis rays* have two zero coordinates (3 rays); *one-zero rays* have exactly one zero ($3n$ rays: normalizing the first nonzero coordinate to 1 leaves one free phase, and there are 3 choices of zero position); *all-nonzero rays* have no zeros (n^2 rays: normalizing the first coordinate to 1 leaves two free phases).

A key structural observation: in $\mathbb{C}^3$, the inner product $\langle v|w \rangle = \sum_k \bar{v}_k w_k$ has at most three nonzero terms. Each nonzero term is $\bar{\zeta}^a \zeta^b = \zeta^{b-a}$, an n th root of unity. Therefore only 1-, 2-, and 3-term vanishing sums of n th roots arise as orthogonality conditions in S_n .

PROOF OF THEOREM 1

Sufficiency

If $6 \mid n$, then $\zeta^{n/6}$ generates the 6th roots of unity as a subgroup of $\langle \zeta \rangle$, so the alphabet for $n = 6$ embeds into that for general n . Since S_6 contains Cabello’s 33-vector KS set [2] and KS-uncolorability is monotone (any superset of an uncolorable set is uncolorable), S_n is KS-uncolorable. $\square$

Necessity: vanishing-sum analysis

We show that if $6 \nmid n$, then S_n admits a valid KS coloring. The proof rests on two lemmas characterizing *short* vanishing sums of roots of unity (2 or 3 terms). We emphasize that longer vanishing sums of n th roots can exist for arbitrary n (see Lam and Leung [6] for the general classification), but only sums of ≤ 3 terms arise as

orthogonality conditions in $\mathbb{C}^3$ (Section 2), so the short-sum characterization suffices.

Lemma 1 (Three-term vanishing sums). *If $u, v, w \in \langle \zeta \rangle$ and $u+v+w=0$, then $3 \mid n$ and $\{u, v, w\} = \{\lambda, \lambda\omega, \lambda\omega^2\}$ for some $\lambda \in \langle \zeta \rangle$, where $\omega = \zeta^{n/3}$. Conversely, $3 \mid n$ suffices.*

Proof. Write $u = \zeta^a$, $v = \zeta^b$, $w = \zeta^c$. If $a = b$, then $\zeta^a(2 + \zeta^{c-a}) = 0$ forces $|\zeta^{c-a}| = 2 \neq 1$; similarly for other coincidences, so a, b, c are distinct. Dividing by ζ^a : $1 + \zeta^p + \zeta^q = 0$ with $p = b-a$, $q = c-a$ both nonzero. Three unit-modulus numbers summing to zero form an equilateral triangle: $|-1 - \zeta^p| = 1$ gives $\cos(2\pi p/n) = -1/2$, hence $p/n \in \{1/3, 2/3\}$, requiring $3 \mid n$. Then $\{1, \zeta^p, \zeta^q\} = \{1, \omega, \omega^2\}$, confirming the classification $\{u, v, w\} = \{\lambda, \lambda\omega, \lambda\omega^2\}$ with $\lambda = \zeta^a$. $\square$

Lemma 2 (Two-term vanishing sums). *The equation $\zeta^a + \zeta^b = 0$ has a solution if and only if $2 \mid n$.*

Proof. $\zeta^a + \zeta^b = 0$ iff $\zeta^{b-a} = -1$ iff $-1 \in \langle \zeta \rangle$ iff $2 \mid n$. $\square$

These lemmas, combined with the zero-pattern classification, determine all orthogonalities in S_n . For each pair of rays, the inner product $\langle v|w \rangle$ is a sum of t roots of unity, where t equals the number of coordinates in which both rays are nonzero:

Ray pair	t	Orth. condition
all-nz vs. all-nz	3	$3 \mid n$
all-nz vs. one-zero	2	$2 \mid n$
one-zero vs. one-zero (same)	2	$2 \mid n$
one-zero vs. one-zero (diff.)	1	never
any vs. axis	0–1	$v_k = 0$

The “diff.” case has $t = 1$ because the two zero coordinates are distinct, so two of three inner-product terms vanish; the surviving term has modulus 1 and cannot vanish. The “any vs. axis” case is a structural zero-pattern constraint, independent of n ; the three axis rays form a triad for all n .

We now treat the three cases where $6 \nmid n$.

Case 1: $3 \nmid n$, $2 \nmid n$

Any triad requires three mutually orthogonal rays spanning $\mathbb{C}^3$. A non-axis triad must contain at least one ray with fewer than two zero coordinates, hence at least one orthogonal pair whose inner product is a 2-term or 3-term sum of roots. Since $3 \nmid n$ forbids 3-term vanishing sums and $2 \nmid n$ forbids 2-term vanishing sums, no such pair can be orthogonal. The sole triad is the axis triad $\{e_1, e_2, e_3\}$, trivially colorable. $\square$

Case 2: $3 \nmid n$, $2 \mid n$

By Lemma 1, no all-nonzero orthogonal pairs exist, so no triad can contain two all-nonzero rays.

Lemma 3 (No all-nonzero ray in any triad when $3 \nmid n$). *If $3 \nmid n$, every triad in S_n contains an axis ray.*

Proof. A triad not containing an axis ray consists of rays each having at most one zero coordinate. Since no two all-nonzero rays are orthogonal ($3 \nmid n$), at most one ray is all-nonzero; the other two are one-zero. Two one-zero rays from *different* planes are never orthogonal (Lemma 5), so both one-zero rays r_1, r_2 share the same zero coordinate k . But then r_1 and r_2 span a subspace of $\{x \in \mathbb{C}^3 : x_k = 0\}$. Since $r_1 \perp r_2$, they form an orthogonal basis for this 2-dimensional subspace, and any vector orthogonal to both must lie in its orthogonal complement: the k -axis. The third ray v is therefore proportional to e_k , contradicting the assumption that v is all-nonzero. A fortiori, three one-zero rays cannot form a triad (the same plane argument applies to any two, and different planes are excluded by Lemma 5). $\square$

Moreover, axis rays are never orthogonal to all-nonzero rays ($e_k \perp v$ requires $v_k = 0$). Since every triad contains an axis ray, no all-nonzero ray appears in any triad. Every triad has the form $\{v_1, v_2, e_k\}$ where v_1, v_2 are one-zero rays in the k th coordinate plane and e_k is the complementary axis ray.

Lemma 4 (Perfect matching). *For even n , the orthogonality graph on one-zero rays in each coordinate plane is a perfect matching.*

Proof. The one-zero rays in the $z = 0$ plane are projectively $(1, \zeta^p, 0)$ for $p \in \{0, \dots, n-1\}$. Two such rays p, q are orthogonal iff $1 + \zeta^{q-p} = 0$, i.e., $q - p \equiv n/2 \pmod{n}$. Each ray has exactly one partner, giving $n/2$ disjoint edges. $\square$

Lemma 5 (Plane independence). *One-zero rays from different coordinate planes are never orthogonal.*

Proof. If v has $v_i = 0$ and w has $w_j = 0$ with $i \neq j$, then $\langle v|w \rangle = \sum_k \bar{v}_k w_k$ has exactly one nonzero term (at the coordinate $k \neq i, j$), which is a product of roots of unity, hence has modulus 1. For example, $\langle (\zeta^a, \zeta^b, 0) | (\zeta^c, 0, \zeta^d) \rangle = \bar{\zeta}^a \zeta^c + 0 + 0 = \zeta^{c-a} \neq 0$. $\square$

We construct an explicit coloring. Assign $e_1 \mapsto 1$, $e_2, e_3 \mapsto 0$. All one-zero rays in the $x=0$ plane receive 0 (their triads with e_1 already have one 1). In each of the $y=0$ and $z=0$ planes, the matched pairs (Lemma 4) are independently 2-colored: one 1, one 0 per pair. All-nonzero rays receive 0.

Verification of (II): each triad $\{v_1, v_2, e_k\}$ has exactly one 1. For $k = 1$: $e_1 = 1$, both partners 0. For $k = 2, 3$: axis is 0, exactly one of the matched pair is 1.

Verification of (I): we check that no two rays assigned 1 are orthogonal. The ray e_1 is orthogonal only to rays with $v_1 = 0$ (i.e., $x=0$ plane rays), which all received 0. Rays assigned 1 in the $y=0$ plane are orthogonal only to their matched partner (assigned 0), to e_2 (assigned 0), and to no ray in the $z=0$ plane (Lemma 5). The same holds symmetrically for $z=0$. All-nonzero rays received 0; since only constraint (I) applies to them (they appear in no triad by the above), and 0-valued rays cannot violate (I), the coloring is valid. $\square$

Case 3: $3 \mid n, 2 \nmid n$

This is the principal case. Let $\omega = e^{2\pi i/3} = \zeta^{n/3}$, a primitive cube root of unity. Three-term cancellations create triads among all-nonzero rays, but we show these triads are pairwise disjoint and the KS constraints decompose. First, since $2 \nmid n$, no 2-term vanishing sums exist, so no all-nonzero ray is orthogonal to any one-zero ray, and no one-zero ray is orthogonal to another one-zero ray. The all-nonzero and zero-bearing sectors are therefore completely decoupled.

Lemma 6 (Projective collapse). *For any all-nonzero ray v in S_n with $3 \mid n$, there are exactly two projectively distinct all-nonzero rays orthogonal to v .*

Proof. Normalize $v = (1, \zeta^p, \zeta^q)$. For all-nonzero $w = (1, \zeta^r, \zeta^s)$,

$$\langle v|w \rangle = 1 + \zeta^{r-p} + \zeta^{s-q}.$$

By Lemma 1, vanishing requires the three terms to form $\{\lambda, \lambda\omega, \lambda\omega^2\}$. Since one term is 1, we need $1 \in \{\lambda, \lambda\omega, \lambda\omega^2\}$, giving $\lambda \in \{1, \omega^2, \omega\}$. All three choices produce the same unordered set $\{1, \omega, \omega^2\}$; what differs is which element is assigned to which of the three terms $1, \zeta^{r-p}, \zeta^{s-q}$. Since the first term is fixed at 1, only two assignments remain: $(\zeta^{r-p}, \zeta^{s-q}) = (\omega, \omega^2)$ or (ω^2, ω) . This gives exactly two projective solutions:

$$\begin{aligned} w_+ &= (1, \omega\zeta^p, \omega^2\zeta^q), \\ w_- &= (1, \omega^2\zeta^p, \omega\zeta^q). \end{aligned}$$

(The three λ -values yield only w_+ and w_- up to projective equivalence.) Since $\omega \neq 1$: $w_+ \not\sim v$, $w_- \not\sim v$, and $w_+ \not\sim w_-$. $\square$

Lemma 7 (Triad uniqueness). *$\{v, w_+, w_-\}$ is a triad, it is the unique triad containing v among all-nonzero rays, and $T_+ : v \mapsto w_+$ generates 3-cycles.*

Proof. Orthogonality: $\langle w_+|w_- \rangle = 1 + \omega^2 \cdot \omega + \omega \cdot \omega^2 = 1 + \omega + \omega^2 = 0$ (using $\bar{\omega} = \omega^2$). Uniqueness: v has exactly two all-nonzero orthogonal neighbors (Lemma 6). The map T_+ acts as $(1, \zeta^p, \zeta^q) \mapsto (1, \omega\zeta^p, \omega^2\zeta^q)$. Applying twice: $T_+^2(v) = (1, \omega^2\zeta^p, \omega\zeta^q) = w_-$. Applying thrice: $T_+^3(v) = (1, \zeta^p, \zeta^q) = v$. So each orbit has size exactly 3. $\square$

Lemma 8 (One-zero isolation). *For odd n with $3 \mid n$, every one-zero ray has exactly one orthogonal neighbor (the complementary axis ray) and participates in no triad.*

Proof. Let v be a one-zero ray with $v_k = 0$. We show v is orthogonal only to the axis ray e_k . Clearly $\langle v|e_k \rangle = 0$ since both nonzero coordinates of v are multiplied by 0. For the converse, consider any other ray $w \not\sim e_k$:

Case (a): w is all-nonzero. Then $\langle v|w \rangle$ has exactly two nonzero terms (at the positions where v is nonzero), each a root of unity. This is a 2-term sum, which vanishes only if $2 \mid n$ (Lemma 2). Since n is odd, $v \not\sim w$.

Case (b): w is one-zero with $w_j = 0, j = k$ (same plane). Then $\langle v|w \rangle$ is again a 2-term sum of roots of unity. Since $2 \nmid n$, $v \not\sim w$.

Case (c): w is one-zero with $w_j = 0, j \neq k$ (different plane). Then exactly one coordinate position has both v and w nonzero, so $\langle v|w \rangle$ is a single root of unity, which cannot vanish. Hence $v \not\sim w$ (this is Lemma 5).

Case (d): w is an axis ray e_j with $j \neq k$. Then $\langle v|e_j \rangle = \bar{v}_j \neq 0$ since $v_j \neq 0$ (the only zero coordinate of v is at position k , not j).

Thus $\deg(v) = 1$ in the orthogonality graph, and v cannot participate in any triad (which requires mutual orthogonality of three rays). $\square$

By Lemma 6, the orthogonality graph on all-nonzero rays is 2-regular. By Lemma 7, the map T_+ generates orbits of size exactly 3, each forming a triangle $\{v, w_+, w_-\}$ with all three edges present. Since each vertex has degree 2 and both neighbors lie in its own triangle, there are no edges between triangles. The n^2 all-nonzero rays thus partition into $n^2/3$ isolated triads, plus the axis triad: $n^2/3 + 1$ triads total.

The all-nonzero triads are pairwise disjoint: no two share a ray (Lemma 7), and no orthogonality edges connect rays in different all-nonzero triads (by Lemma 6, each all-nonzero ray has exactly two orthogonal neighbors, both in its own triad). Since $2 \nmid n$, no all-nonzero ray is orthogonal to any one-zero ray. Moreover, no all-nonzero ray is orthogonal to any axis ray: $\langle e_k|w \rangle = \bar{w}_k \neq 0$ when all coordinates of w are nonzero. By Lemma 8, one-zero rays have degree 1 (connected only to their complementary axis ray) and participate in no triad. Note that the axis triad is *not* isolated in the orthogonality graph—each axis ray e_k is orthogonal to all one-zero rays with $v_k = 0$ —but these pendant edges impose only constraint (I), not (II), since one-zero rays appear in no triad. The all-nonzero triads, the axis triad, and the pendant one-zero rays are therefore mutually decoupled.

The KS coloring constraints decompose into independent subproblems: assign one 1 per all-nonzero triad independently, assign $e_1 \mapsto 1$ in the axis triad, and assign 0 to all one-zero rays. Every triad receives exactly one 1. No two rays assigned 1 are orthogonal: within each all-nonzero triad, exactly one ray is 1 and the other two

are 0; the single axis ray assigned 1 (e_1) is not orthogonal to any all-nonzero ray; and all one-zero rays received 0. $\square$

DISCUSSION

The factorization $6 = 2 \times 3$ in Theorem 1 reflects two independent algebraic mechanisms, each necessary for KS-uncolorability:

Phase cancellation ($3 \mid n$): the identity $1 + \omega + \omega^2 = 0$ creates orthogonal triads among all-nonzero rays. Without it, no all-nonzero pair is orthogonal, and triads are restricted to zero-coordinate configurations.

Interlocking ($2 \mid n$): the element $-1 \in \langle \zeta \rangle$ creates two-term cancellations that link triads together. Without it, the projective collapse of Lemma 6 ensures every all-nonzero triad is pairwise disjoint and disconnected from every other, making the coloring problem decomposable.

The factorization $6 = 2 \times 3$ is suggestively consistent with—but logically independent of—the ring-theoretic result of Cortez, Morales, and Reyes [3], who proved that $\mathbb{Z}[1/6]$ is the minimal subring of $\mathbb{Q}$ over which KS projection matrices can be realized in dimension 3. Their result concerns realizability of $P_v = vv^\dagger / \|v\|^2$ over subrings, a distinct question from ours about colorability of ray sets generated by a restricted coordinate alphabet. The two perspectives agree that primes 2 and 3 are both necessary, but the mechanisms differ: their result is about ring invertibility of squared norms, whereas ours is about vanishing sums among roots of unity. The present proof

supplies an explicit constructive complement: projective collapse isolates triads when $2 \nmid n$ (Case 3), and plane matching yields an explicit coloring when $3 \nmid n$ (Case 2).

More broadly, the proof technique—classifying orthogonalities by the length of the underlying vanishing sum, then analyzing the resulting constraint graph—may extend to other algebraic families of KS constructions. Whether the projective collapse of Lemma 6, in which group-theoretic symmetry forces orthogonal neighbors to coalesce, occurs in other algebraic settings (e.g., Gaussian or quadratic integer alphabets) remains an open question.

The author acknowledges the use of Claude (Anthropic) for computational assistance. The proof is self-contained; independent numerical cross-checks using PySAT (Glucose4) verified the coloring constructions for all $n \leq 100$. Code is available at <https://github.com/michaelkernaghan/contextuality>.

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